Module 4 | Numerical Solution of Ordinary Differential Equations

The next module in the course addresses techniques for approximating the solution of an *ordinary differential equation*, and *initial value problems* in particular.

Differential equations play the dominant role in applied mathematics. Most systems or phenomena in nearly every corner of physics, chemistry, biology, engineering, finance, and beyond are concerned with how a quantity changes with respect to another variable, often the time variable (we will denote this as t). The mathematical description of the model involves the rate of change of a variable with respect to time, which can be mathematically represented using differential equations. A solution of the model produces a state of the system at certain points in time: the past, present or future.

However, most of these mathematical models of real-world situations are too complicated to have an analytical solution. For many practical problems involving nonlinearities, one cannot write down an explicit closed-form solution; however, approximate solutions to these equations are easier to obtain via numerical methods. These are often derived from the techniques of interpolation, approximation, and quadrature studied earlier in the course.

§4.1 Review of Ordinary Differential Equations

A **differential equation** is any equation containing the derivative of one or more dependent variables, with respect to one or more independent variables. A solution to such an equation is a function (not a number) which, when substituted into the original equation, creates a true statement.

In most applications, we do not have a simple scalar equation, but rather a system of equations describing the coupled dynamics of several variables. Such situations give rise to vector-valued functions $y(t) \in \mathbb{R}^n$. But for the sake of simplicity, we will also assume that $n = 1$. In particular, the initial value problem we will be working with is

Given: $y'(t) = f(t, y(t))$, with $y(t_0) = y_0$, where $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$

Determine: $y(t) = y_0 + \displaystyle\int_{t_0}^{t} f(t, y(t))\,\mathrm{d}t$ for all $t_0 \le t \le \mathrm{T}$

We will not cover analytical methods for solving differential equations in this course; you are welcome to take a DE class to learn more in that area. Instead, let's focus on how we might be able to find a numerical approximation of the solution.

To that end, we start by discretizing the time variable and look for our solution at a discrete number of points. More precisely, we approximate our true solution at these points. The effectiveness of the algorithm will be measured by its ability to approximate such solutions on a large range of problems. What if we want to find the solution between two points for which we have a numerical solution? We will just use Interpolation!

4.1.1 Lipschitz Condition

Since we will be approximating a continuous solution with a discrete amount of data, on a computer, we should always be concerned with round-off error. More precisely, we need to know whether small changes in the statement of the problem introduce correspondingly small changes in the solution.

The following notion of continuity is helpful for this purpose.

Definition 4.1.25

A function $f(t,y) \in C^0(\mathbb{R} \times [t_0, T])$ is said to satisfy a Lipschitz condition in the variable y if there exists some constant $L > 0$ such that,

$$\left|f(t,y_2) - f(t,y_1)\right| \leq L\left|y_2 - y_1\right|, \qquad \text{for all } y_1, y_2 \in \mathbb{R} \text{ and } t \in [t_0, T]$$

The constant L is called a Lipschitz constant for f.

Question 50.

If f and $\frac{\partial f}{\partial y}$ are continuous at a given point, then show that f is Lipschitz in y in a neighborhood.

□

Question 51.

Define a mapping Φ from $C^0[0,T]$ to itself defined by

$$\Phi(y)(t) = y_0 + \int_0^t f(t, y(t))\,dt$$

Show that if f is Lipschitz w.r.t. y with Lipschitz constant L, then Φ is Lipshitz and a contraction mapping if $T < 1/L$.

Consequently, show that if there is a fixed point for Φ, then it is unique on $[0, 1/L]$.

□

The above result, along with the Contraction Mapping Theorem can be used to show that under certain conditions, the solution to an IVP exists and is unique.

4.1.2 Existence and Uniqueness Theorem

We are now ready to describe a class of initial-value problems that can be solved numerically.

Theorem 4.1.26: Picard's Theorem

Consider the rectangle $D = [t_0, T] \times [y_0 - c, y_0 + c]$ for some fixed $c > 0$ and let $f \in C^0(D)$.

Furthermore, suppose $\left|f(t, y_0)\right| \leq K$ for all $t \in [t_0, T]$, and f is Lipschitz in y with Lipschitz constant $L > 0$ on D.

Finally, suppose that

$$c \geq \frac{K}{L}\left(e^{L(T-t_0)} - 1\right).$$

(i.e., the box D must be sufficiently large to compensate for large values of K and L.)

Then there exists a unique $y \in C^1[0, T]$ *such that*

$$y(t_0) = y_0, \quad y'(t) = f(t, y)$$

for all $t \in [0, T]$, *and* $\left|y(t) - y_0\right| \leq c$ *for all* $t \in [t_0, T]$.

In simpler words, there is a unique C^1 solution to the IVP that stays within the rectangle D for all $t \in [t_0, T]$ and as t increases the solution will 'exit' from the right-side of the rectangle, not the top or bottom.

Note: In particular, if $c = \infty$, and the functions $f(t, y)$ and $\dfrac{\partial f}{\partial y}$ are continuous, then the IVP has a unique solution for all t.

4.1.3 Well-posed Problems

Once we have answered the question of Existence and Uniqueness, we still need to determine whether small changes in the statement of a particular problem will induce small changes in the solution. The following definition characterizes problems for which numerical solution is feasible.

Definition 4.1.27

The initial value problem is said to be **well-posed** if a unique solution $y(t)$ exists, and depends continuously on the data y_{t_0} and $f(t, y)$. In other words,

There exist constants $\varepsilon_0 > 0$ and $k > 0$ such that for any ε, with $\varepsilon_0 > \varepsilon > 0$, whenever $\delta(t)$ is continuous with $|\delta(t)| < \varepsilon$ for all t in $[t_0, T]$, and when $|\delta_0| < \varepsilon$, the initial-value problem

$$\frac{dz}{dt} = f(t, z) + \delta(t), \quad t_0 \leq t \leq T \quad z(t_0) = y_0 + \delta_0,$$

has a unique solution $z(t)$ that satisfies

$$|z(t) - y(t)| < k\varepsilon \quad \text{for all } t \text{ in } [t_0, T].$$

Fortunately, Picard's theorem further guarantees that the IVP is well-posed.

Theorem 4.1.28

Suppose $D = \{(t, y) \mid t_0 \leq t \leq T$ *and* $-\infty < y < \infty\}$. *If* f *is continuous and satisfies a Lipschitz condition in the variable* y *on the set* D, *then the initial-value problem*

$$\frac{dy}{dt} = f(t, y), \quad t_0 \leq t \leq T, \quad y(t_0) = y_0$$

is well-posed.